

Product Carbon Footprint Study Report

Product Name: tablet production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: tablet production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 Item(s) tablet production

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 Item(s) tablet production.

2. System Boundary

consumer electronics production, mobile device, tablet, in China.

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of tablet production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **82.164 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

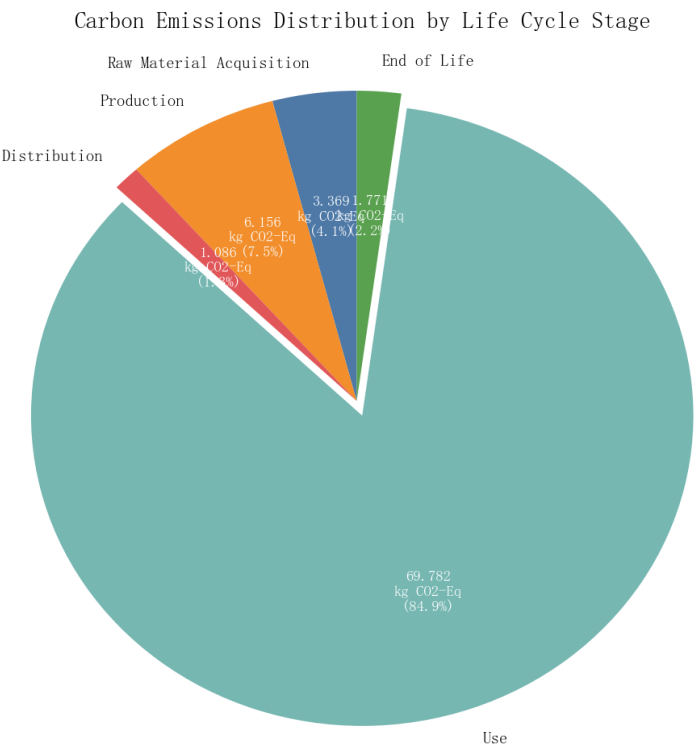


Table 1: Carbon Emissions by Life Cycle Stage

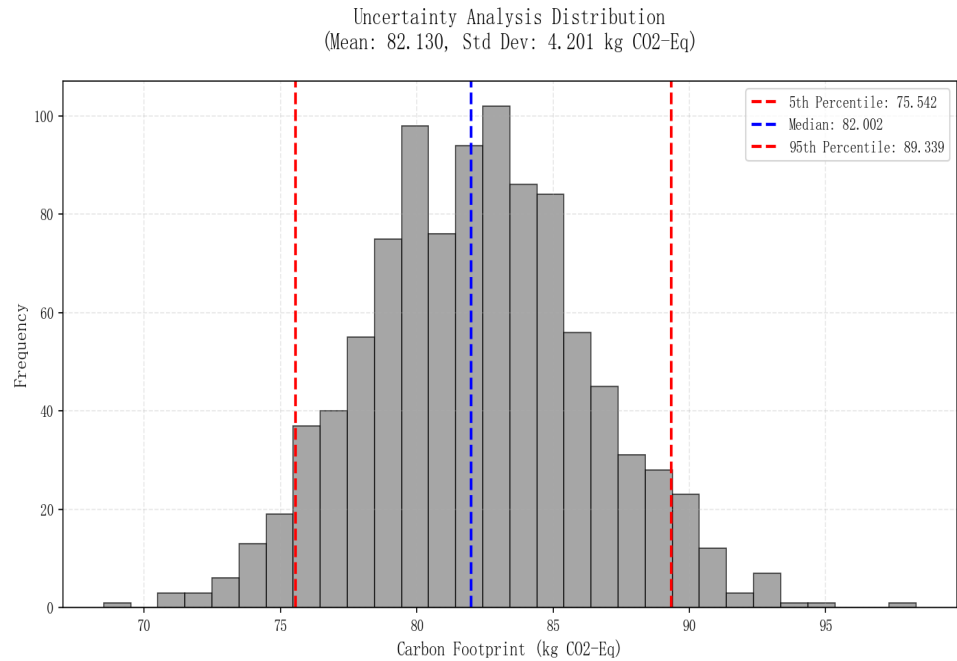
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	3.369	4.10
Production	6.156	7.49
Distribution	1.086	1.32

Use	69.782	84.93
End of Life	1.771	2.16
Total	82.164	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	82.130
Standard Deviation	4.201
Coefficient of Variation (CV)	5.11%
Median	82.002
5th Percentile	75.542
95th Percentile	89.339

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the tablet is 82.164 kg CO₂-Eq, representing a moderate to high impact level for a consumer electronic device over its lifecycle. This value aligns with typical ranges for tablets when accounting for global electricity mixes and average usage durations. The lifecycle contributions reveal a highly skewed distribution, with the use phase dominating at 84.9% (69.782 kg CO₂-Eq), followed distantly by production (7.5%) and raw materials (4.1%). Distribution and end-of-life stages contribute minimally at 1.3% and 2.2%, respectively.

This pattern indicates that operational energy consumption during the use phase constitutes the primary emission driver, likely stemming from electricity demand for charging and processing over an extended product lifetime. Secondary sources in manufacturing and material extraction reflect upstream energy and resource intensities but remain overshadowed by downstream use. Overall, the results highlight the need to prioritize use-phase interventions for meaningful decarbonization in tablet LCAs.

2. Statement of Assumptions and Limitations

Key assumptions in this study include standardized average usage patterns (e.g., daily screen-on time and charging frequency), a representative global or regional electricity grid mix for the use phase, and a typical product lifespan of several years. Data sources are presumed to draw from established databases such as Ecoinvent or GaBi, with proxy datasets applied where specific tablet components lack direct inventories. These assumptions facilitate modeling but may not capture regional variations in consumer behavior or rapid technological shifts in device efficiency.

Limitations arise from data quality issues, including potential outdated emission factors for electronics and incomplete coverage of rare earth elements or supply chain specifics. System boundary restrictions encompass a cradle-to-grave scope limited to the five defined stages, excluding indirect impacts such as infrastructure for data centers, consumer travel for retail, or rebound effects from increased device usage. This boundary ensures focus but may underrepresent broader societal emissions associated with digital consumption.

3. Suggestions for Improvement

- Target the use phase by designing tablets with lower-power processors and optimized battery management to reduce electricity demand by at least 20-30% over the product lifetime.
- Promote renewable energy integration through user incentives or bundled solar chargers to shift the 84.9% use-phase emissions toward lower-carbon sources.
- Enhance raw material and production efficiency via increased recycled content in casings and components, potentially cutting the combined 11.6% contribution from these stages.
- Implement take-back programs to improve end-of-life recycling rates, minimizing the 2.2% disposal impact while recovering valuable materials.
- Improve data quality by conducting primary data collection from suppliers and performing sensitivity analyses on electricity mix assumptions to refine future LCA iterations.

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